

The Course

Synthetic Aperture Radar (SAR)

Part 3: SAR Interferometry

Prepared by DLR-HR's Pol-InSAR Team

German Aerospace Center (DLR), Microwaves & Radar Institute (HR), Pol-InSAR Research Group

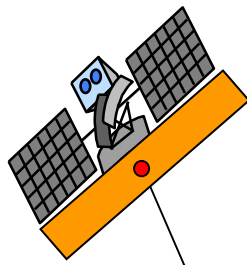
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ESAMAAP

EEBI  **MASS**

SAR Interferometry

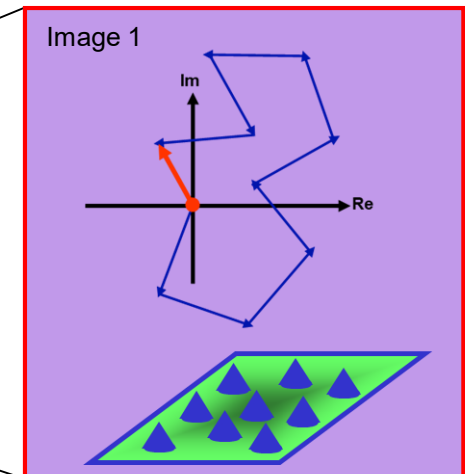
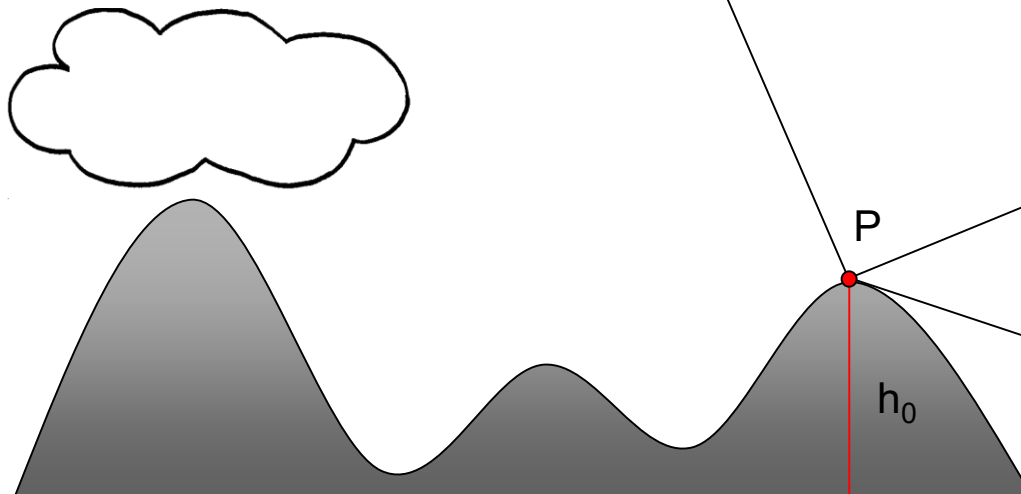


Signal from resolution cell P in Image 1: $i_1 = |i_1| \exp[-i(2\frac{2\pi}{\lambda}R_1) + \varphi_{s1}]$

Phase: $\varphi_1 = \arg(i_1) = \left(2\frac{2\pi}{\lambda}R_1\right) + \varphi_{s1}$

Term 1: Deterministic - proportional to the range distance R_1 of P

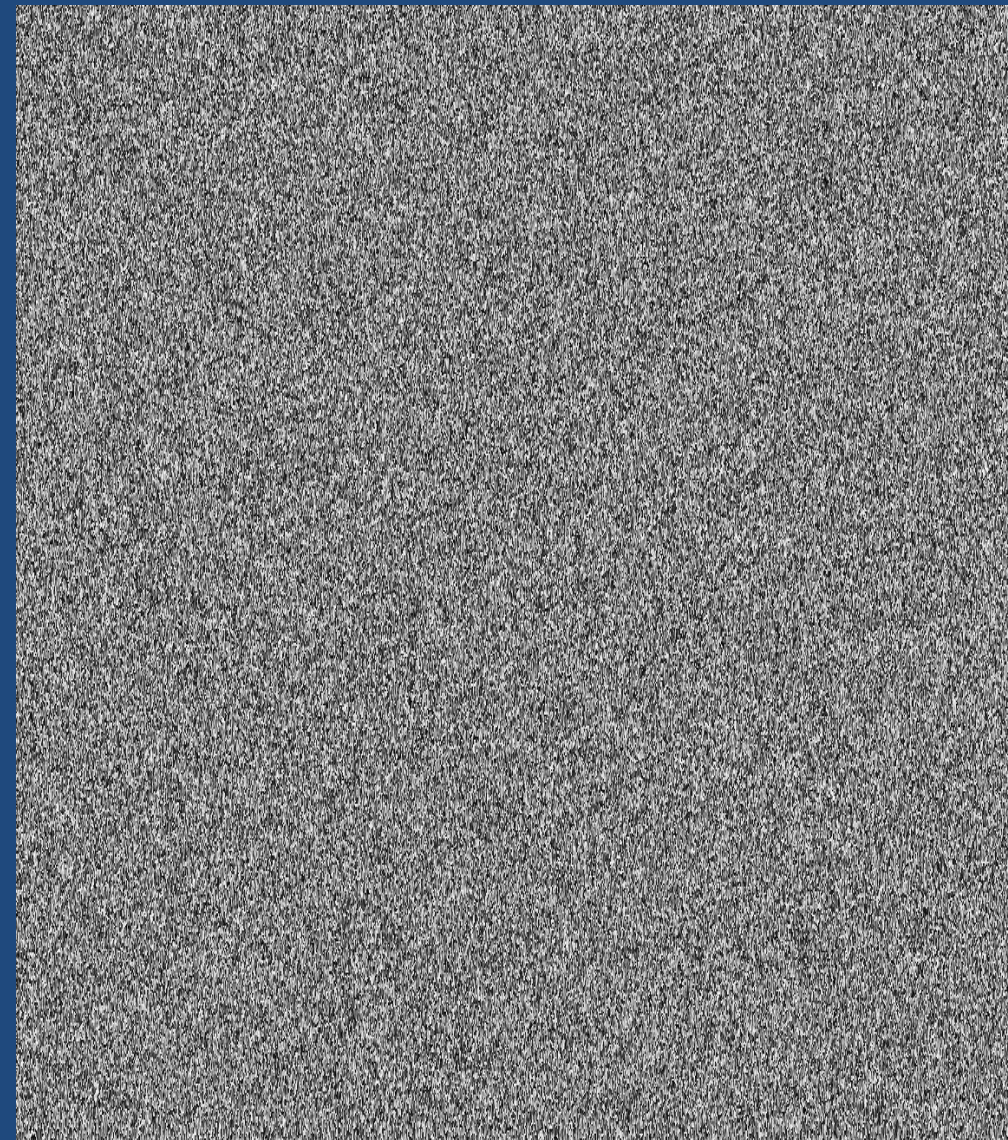
Term 2: Stochastic - induced by the scatterer (Speckle)



ERS – Bachu / China ~ 100 km × 80 km

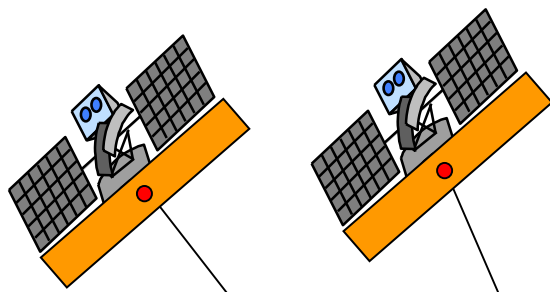


Amplitude Image 1



Phase Image 1

SAR Interferometry



$$R_2 = R_1 + \Delta R$$

R_1

Signal from resolution cell P in Image 1: $i_1 = |i_1| \exp[-i(2\frac{2\pi}{\lambda}R_1) + \varphi_{S1}]$

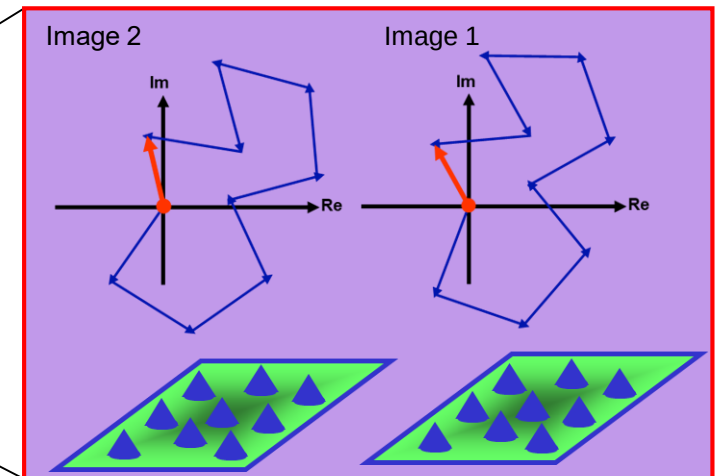
Phase:

$$\varphi_1 = \arg(i_1) = \left(2\frac{2\pi}{\lambda}R_1\right) + \varphi_{S1}$$

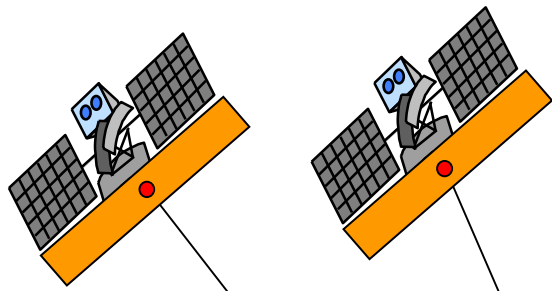
Signal from resolution cell P in Image 2: $i_2 = |i_2| \exp[-i(2\frac{2\pi}{\lambda}R_2) + \varphi_{S2}]$

Phase:

$$\varphi_2 = \arg(i_2) = \left(2\frac{2\pi}{\lambda}R_2\right) + \varphi_{S2}$$



SAR Interferometry



$$R_2 = R_1 + \Delta R$$

R_1

Signal from resolution cell P in Image 1: $i_1 = |i_1| \exp[-i(2\frac{2\pi}{\lambda}R_1) + \varphi_{S1}]$

Phase:

$$\varphi_1 = \arg(i_1) = \left(2\frac{2\pi}{\lambda}R_1\right) + \varphi_{S1}$$

Signal from resolution cell P in Image 2: $i_2 = |i_2| \exp[-i(2\frac{2\pi}{\lambda}R_2) + \varphi_{S2}]$

Phase:

$$\varphi_2 = \arg(i_2) = \left(2\frac{2\pi}{\lambda}R_2\right) + \varphi_{S2}$$

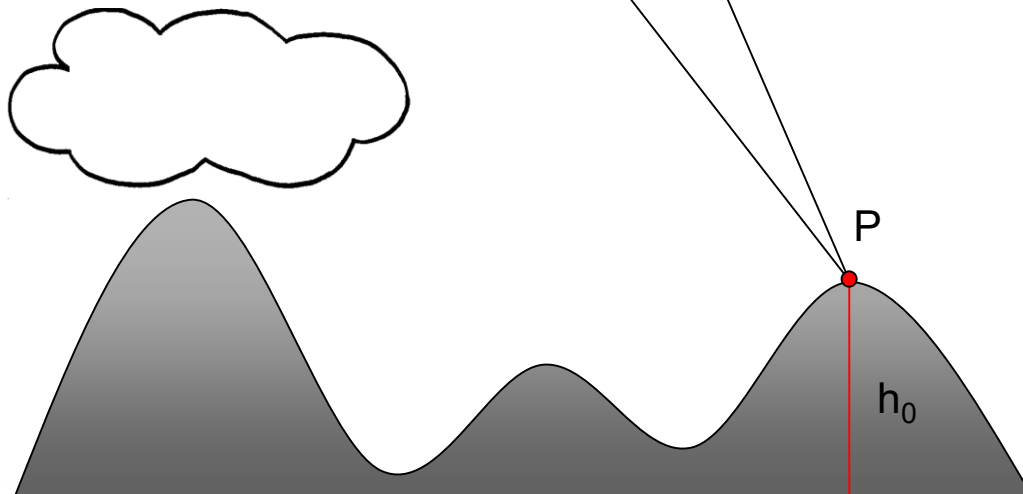


Assuming $\varphi_{S1} \approx \varphi_{S2} !!!$

Interferogram: $i_1 i_2^* = |i_1 i_2^*| \exp[-i(2\frac{2\pi}{\lambda}\Delta R)]$

$$\text{Phase: } \varphi_{\text{Int}} = \arctan\left(\frac{\text{Im}\{i_1 i_2^*\}}{\text{Re}\{i_1 i_2^*\}}\right)\Bigg|_{-\pi}^{+\pi} = \left[2\frac{2\pi}{\lambda}\Delta R\right]$$

Deterministic !!!



ERS – Bachu / China ~ 100 km × 80 km



Amplitude Image 1

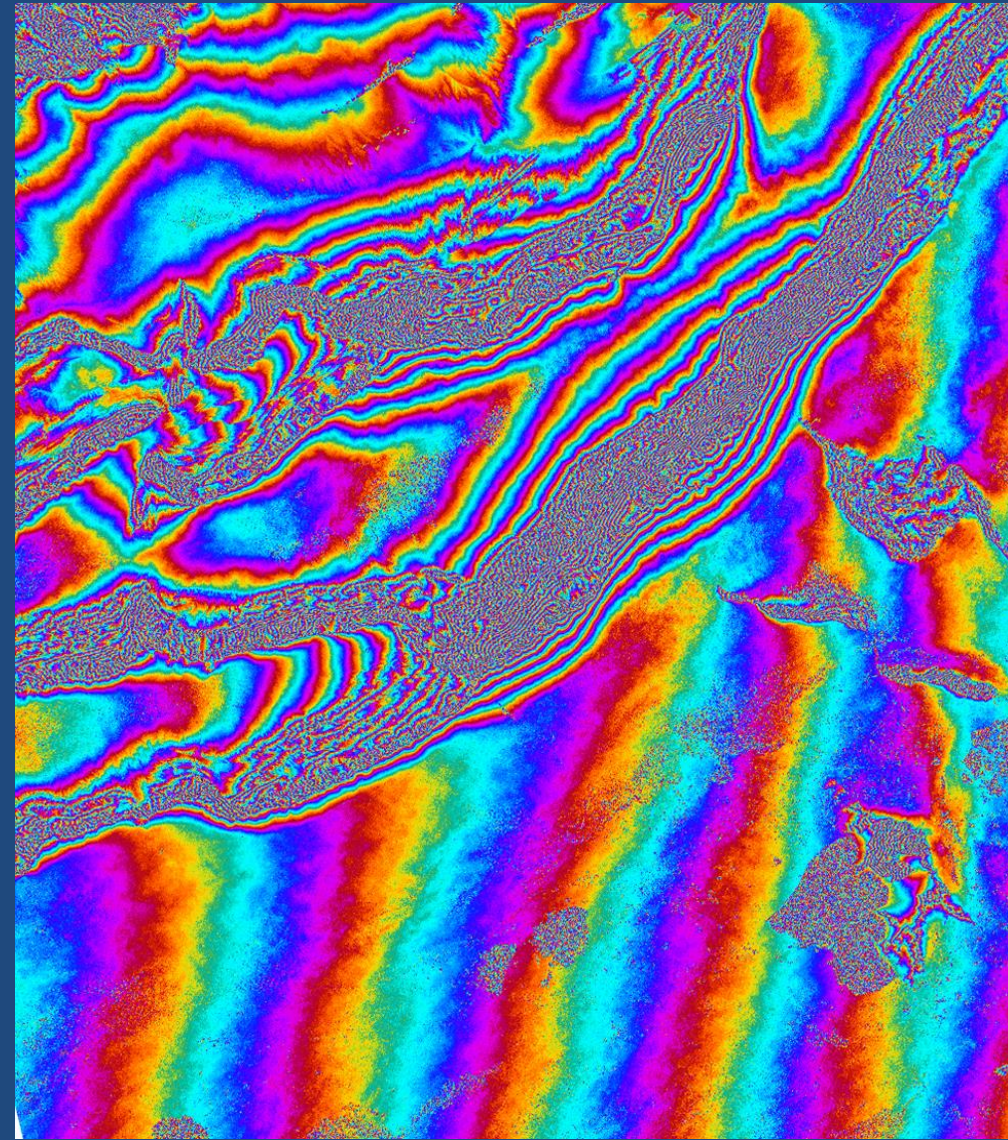


Amplitude Image 2

ERS – Bachu / China ~ 100 km × 80 km

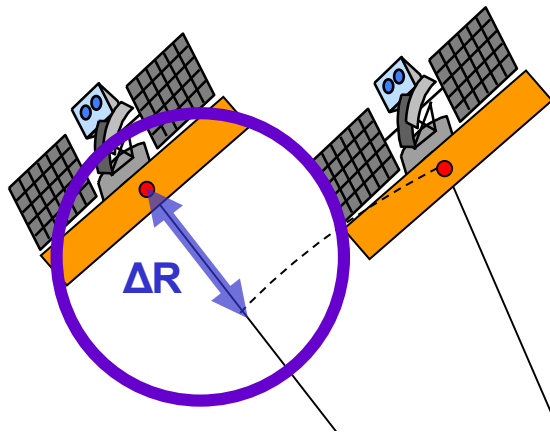


Interferogram Amplitude



Interferometric Phase Image

SAR Interferometry



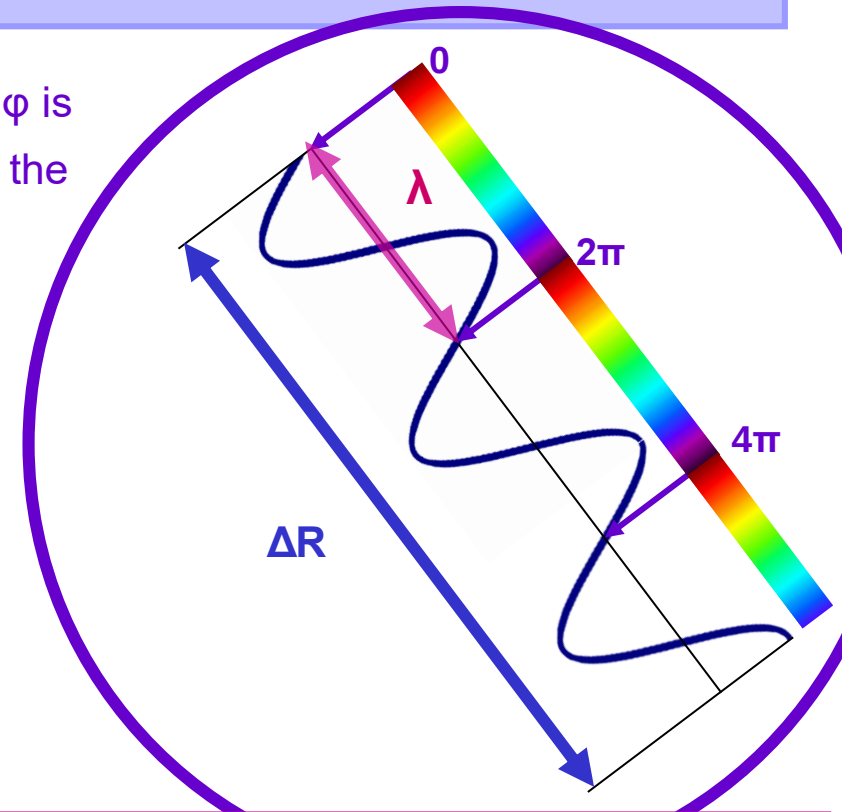
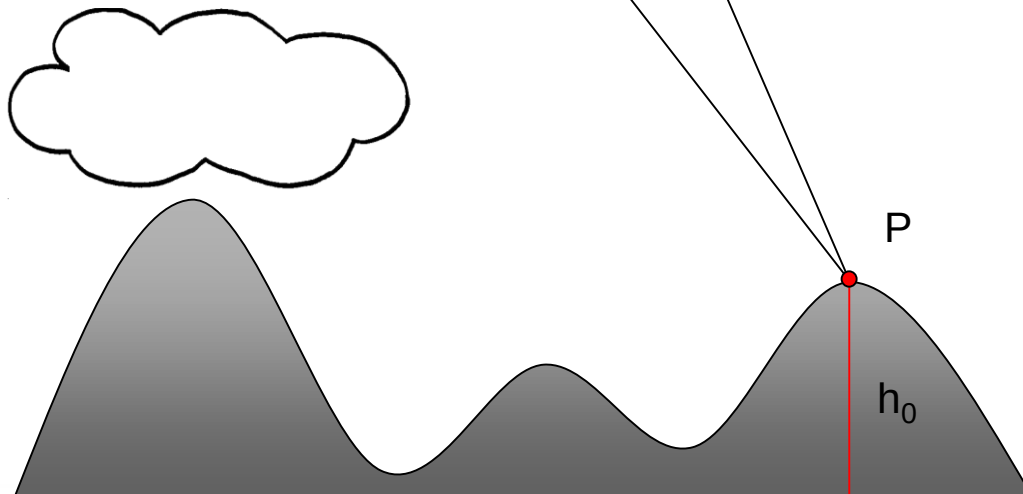
Interferometric Phase: $\varphi = 2\frac{2\pi}{\lambda} \Delta R$

Parallax Measurement

$$R_2 = R_1 + \Delta R$$

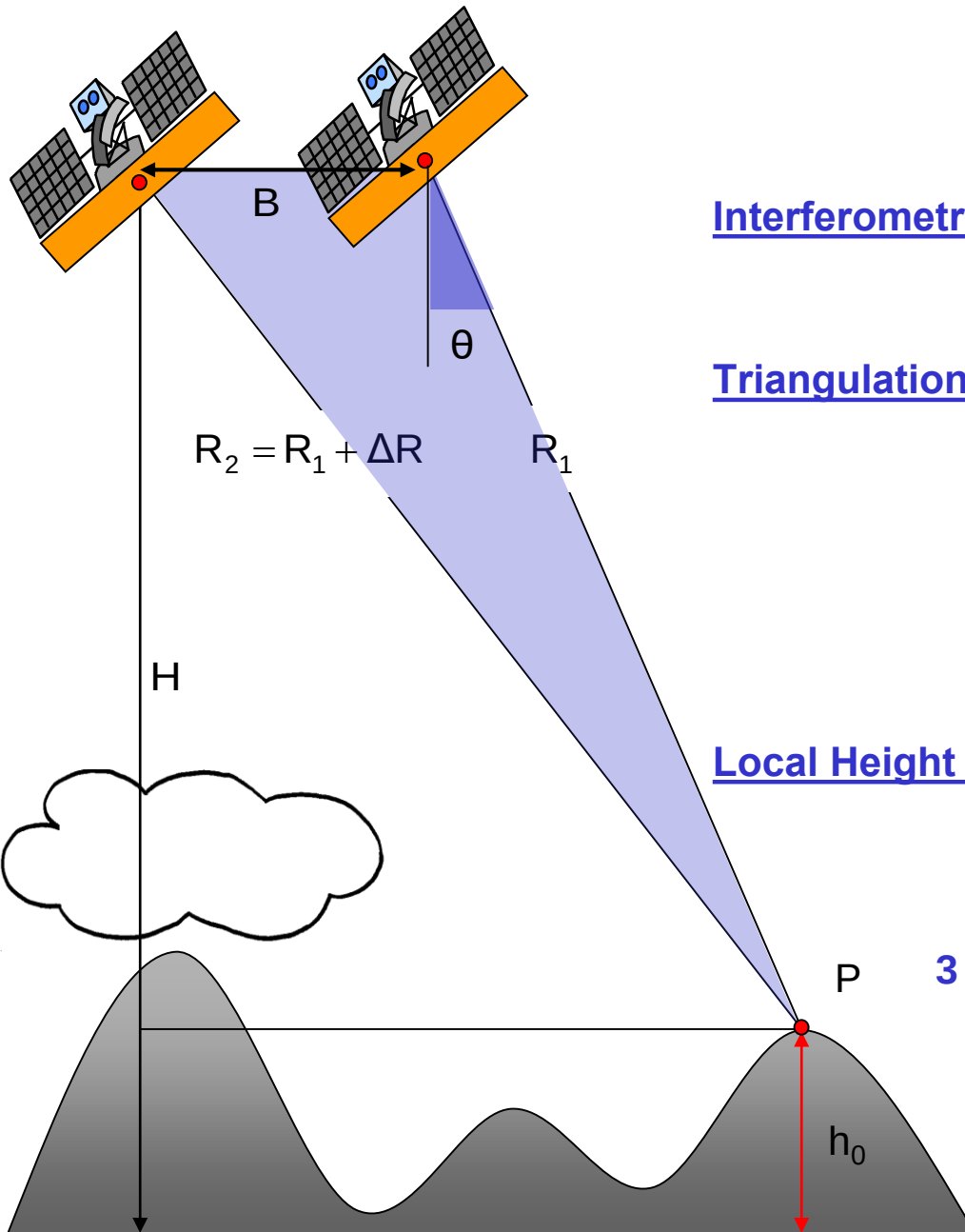
R_1

The interferometric phase φ is another way to measure the (relative) distance Δr in sub-wavelength scales!



Phase measurements in interferometric systems can be made with a degree level accuracy. At radar wavelengths of 1-90cm (Ku to P-band) this corresponds to sub-millimeter accuracy !!!

DEM Generation



Interferometric Phase (1): $\varphi = 2 \frac{2\pi}{\lambda} \Delta R + 2\pi N \quad N = 0, \pm 1, \pm 2$

Triangulation (2): $(R_1 + \Delta R)^2 = R_1^2 + B^2 - 2R_1B \cos(\pi/2 + \theta) \rightarrow$

$$\rightarrow \sin(\theta) = \frac{(R_1 + \Delta R)^2 - R_1^2 - B^2}{2R_1B}$$

Local Height (3): $h_0 = H - (R_1 + \Delta R) \cos(\theta)$



3 non-linear Equations for 3 Unknowns (h_0 , θ , ΔR)

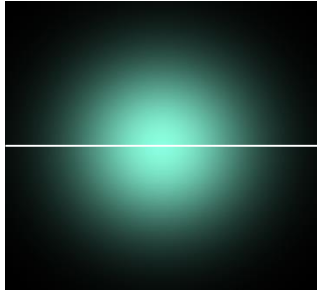
B ... Spatial baseline and

R_1 ... Range distance in Image 1 are known

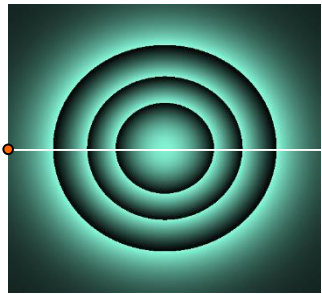
Critical is the fact that the interferometric phase φ is initially measured modulo 2π ► **Phase Unwrapping**

Phase Unwrapping

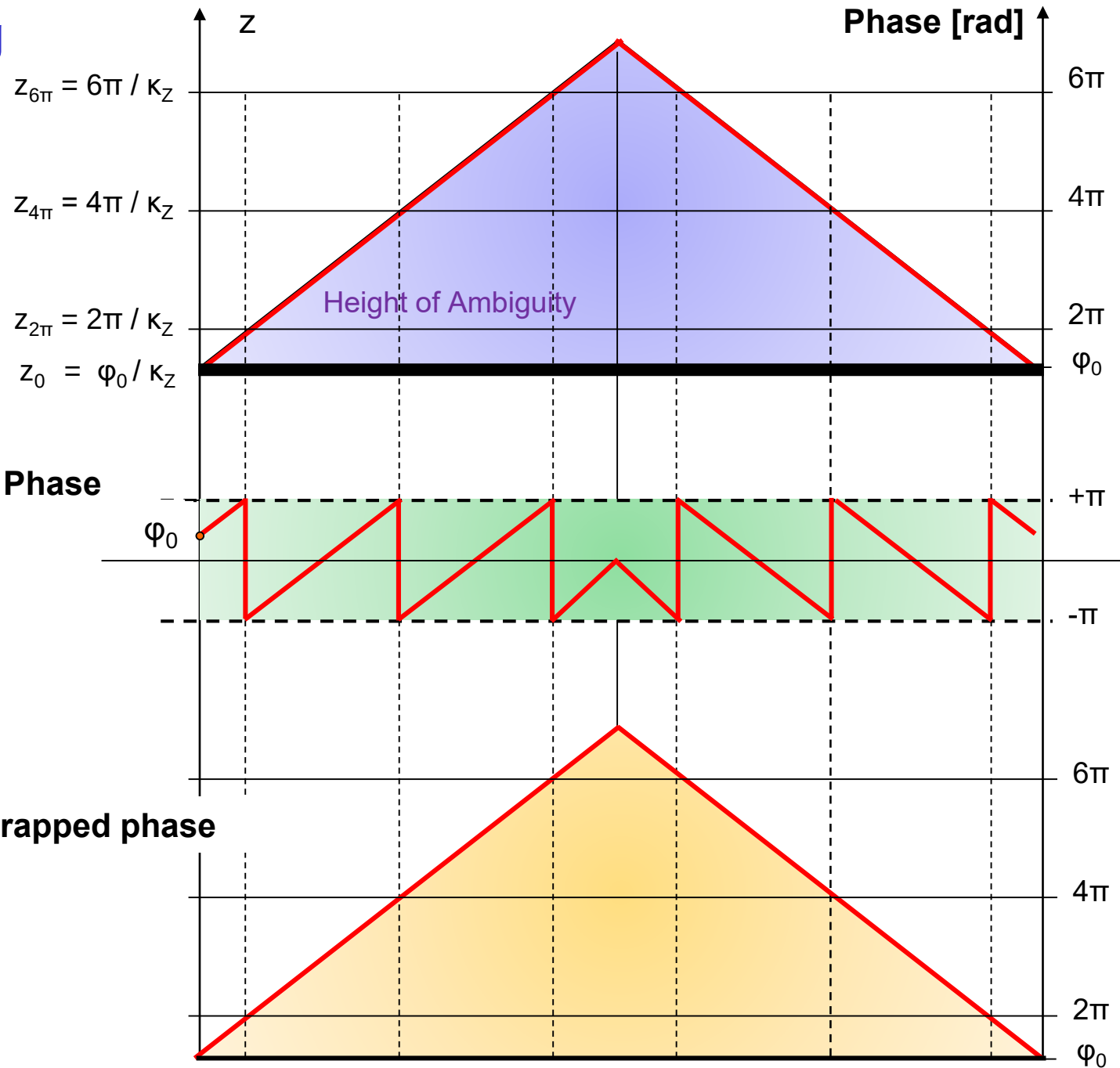
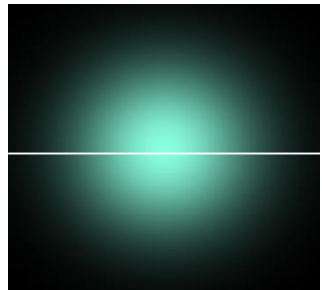
Original Signal



Measured Signal: Wrapped Phase



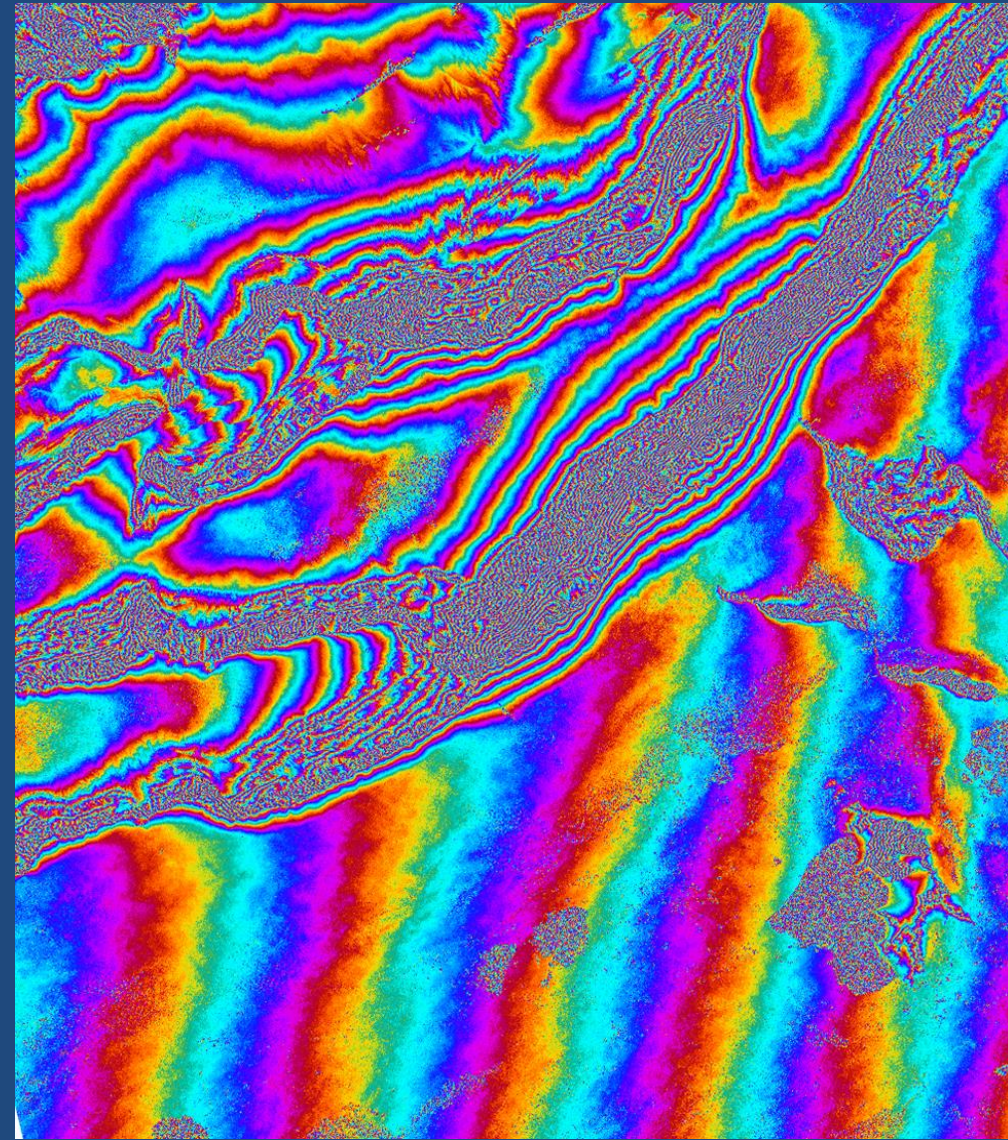
Reconstructed Signal: Unwrapped phase



ERS – Bachu / China ~ 100 km × 80 km



Amplitude Image 1

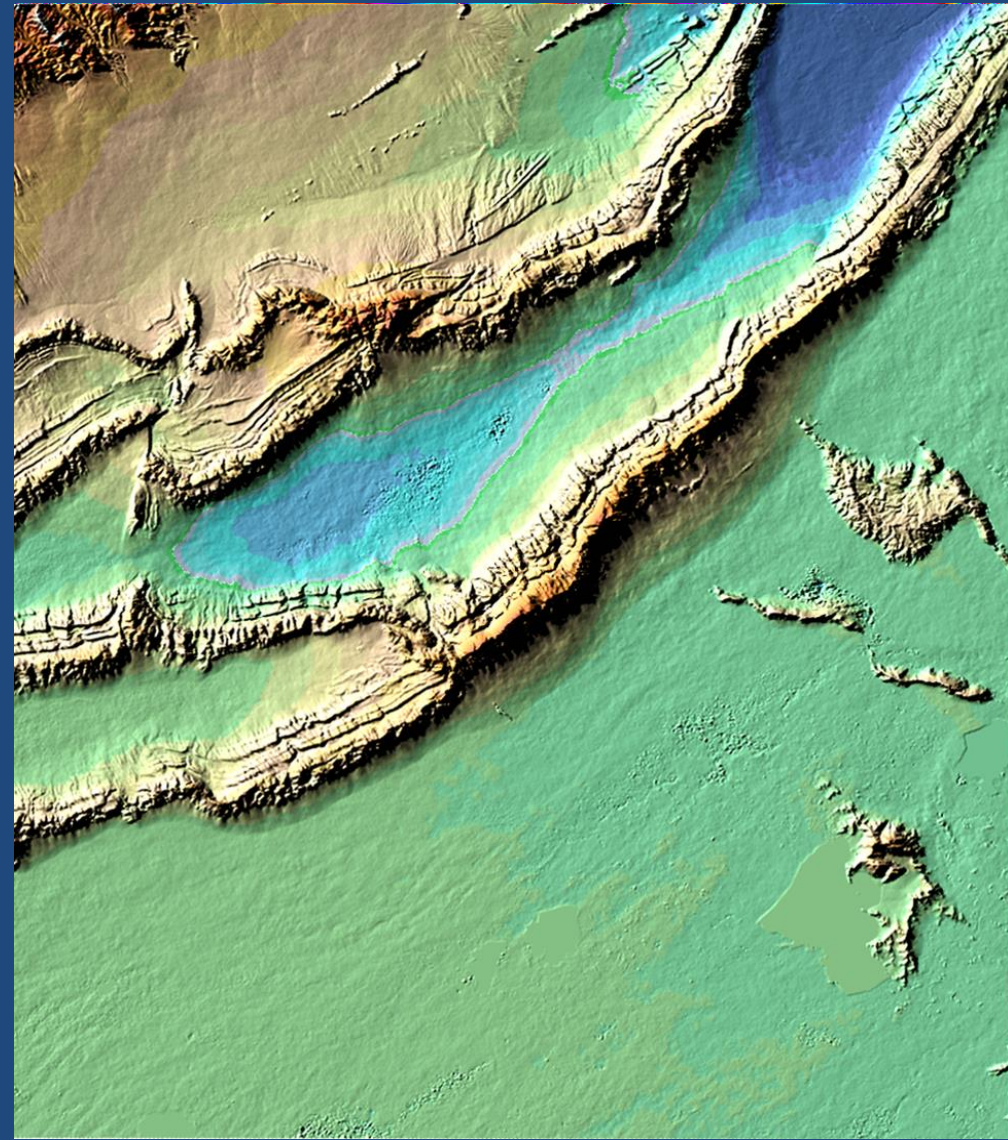


Interferometric Phase Image

ERS – Bachu / China ~ 100 km × 80 km

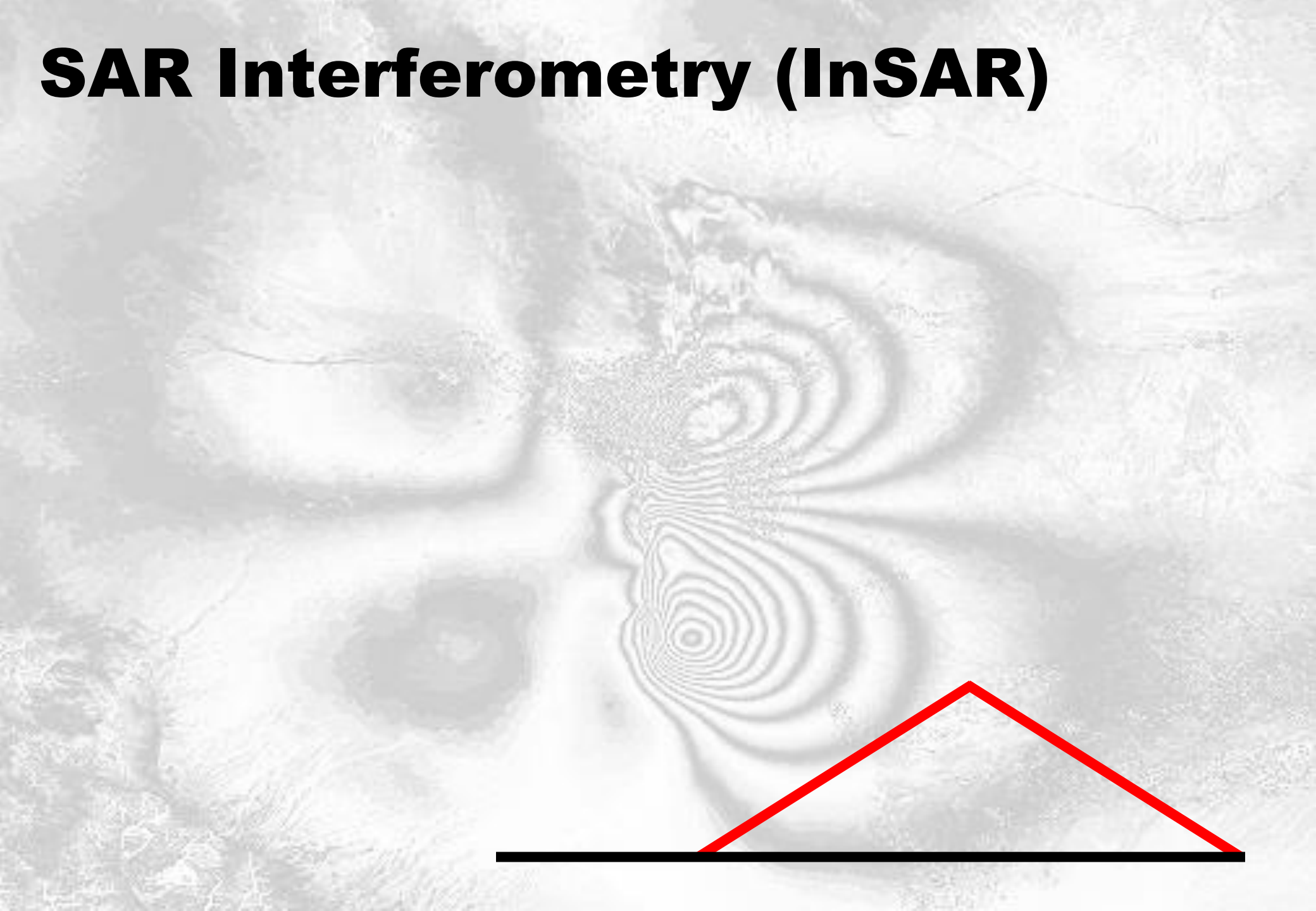


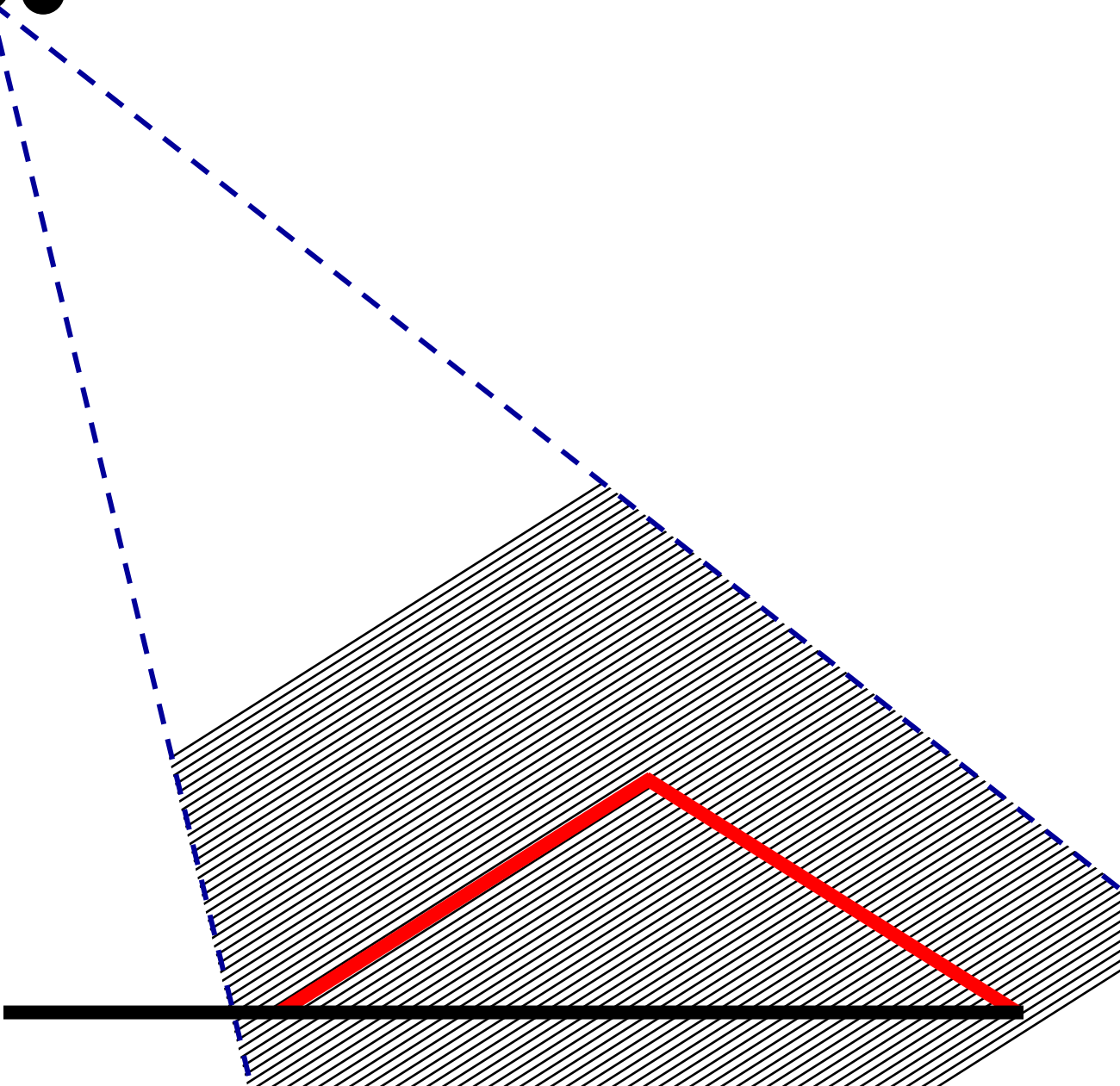
Amplitude Image

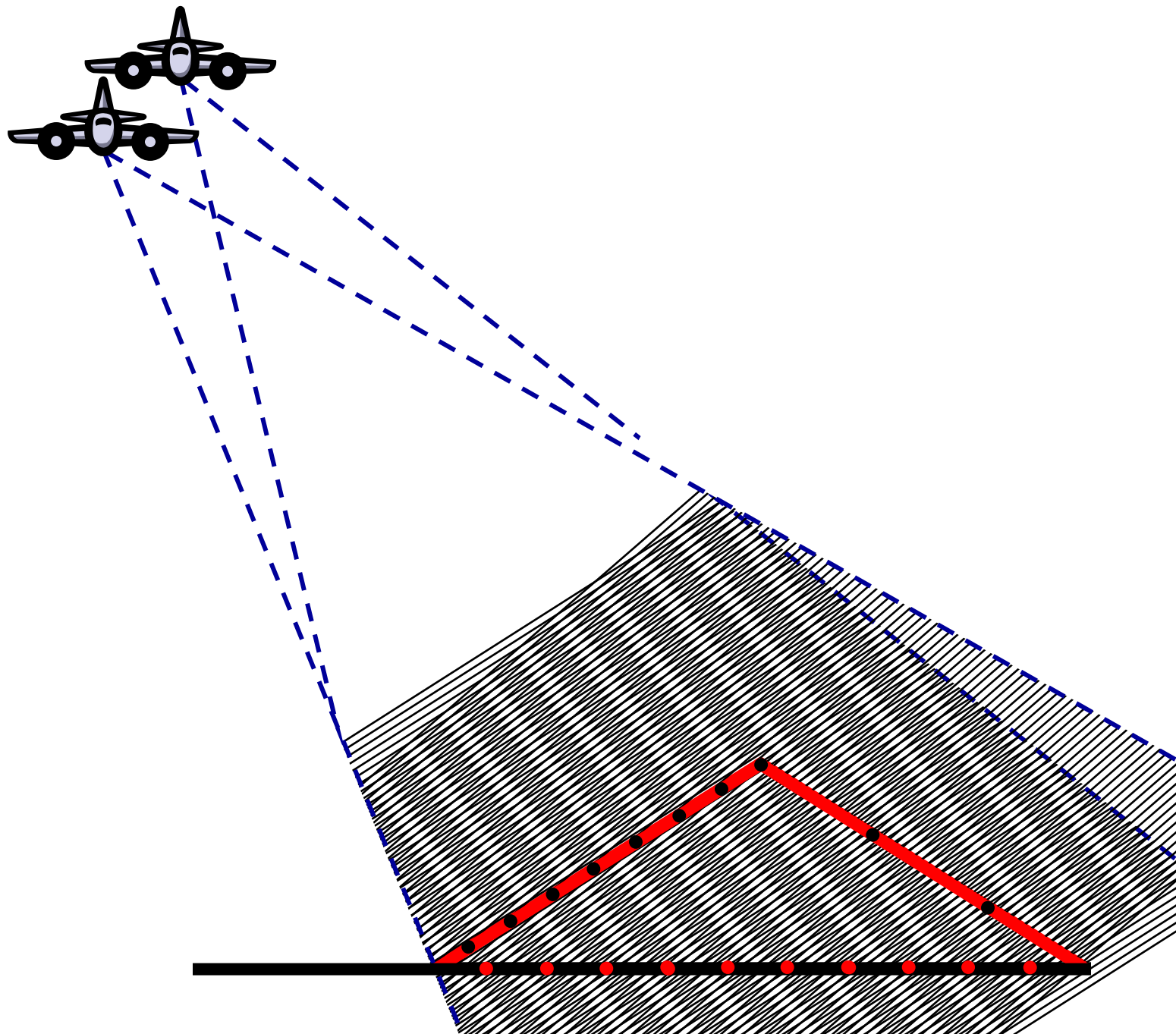


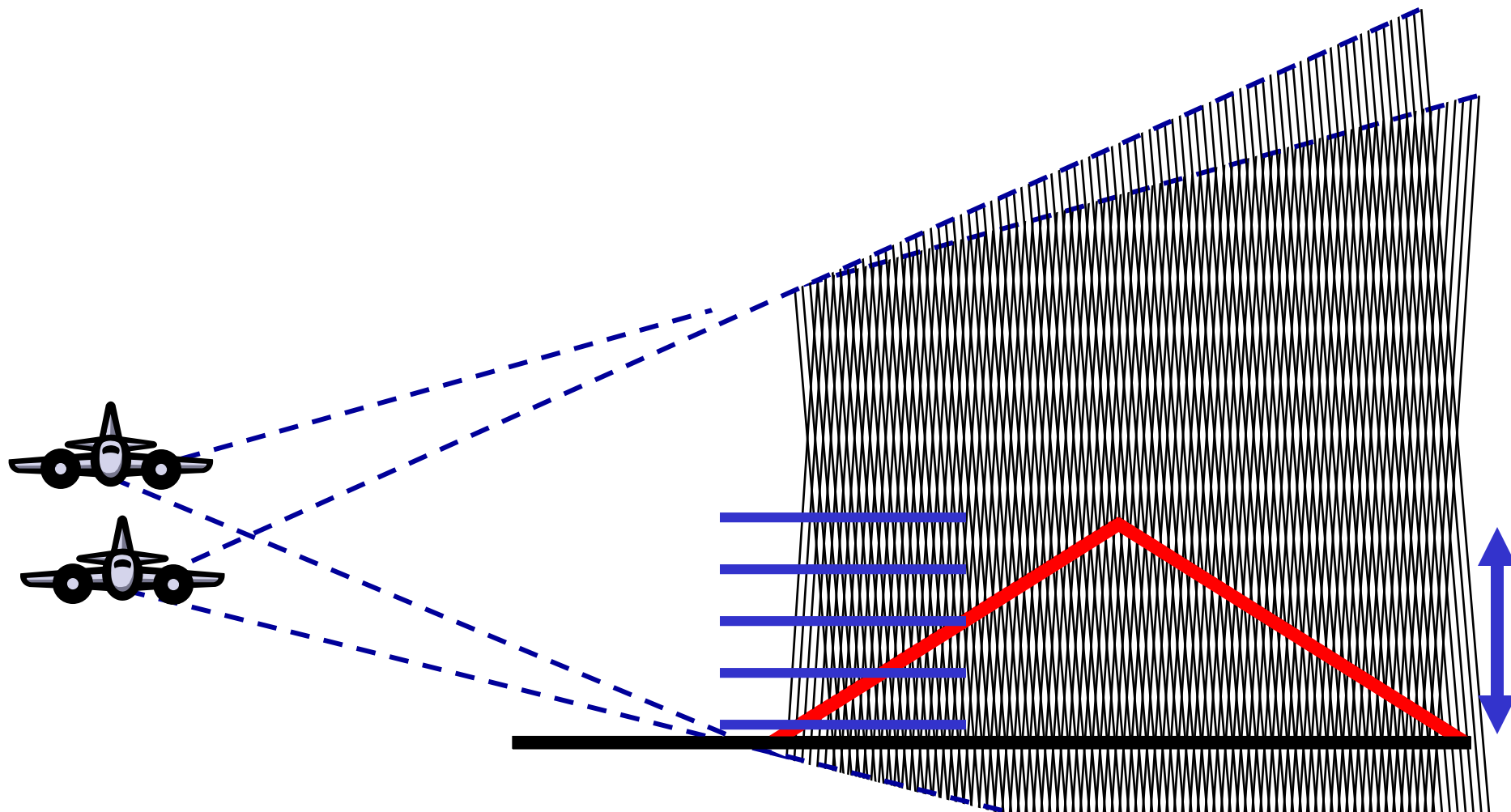
Digital Elevation Model with false colors

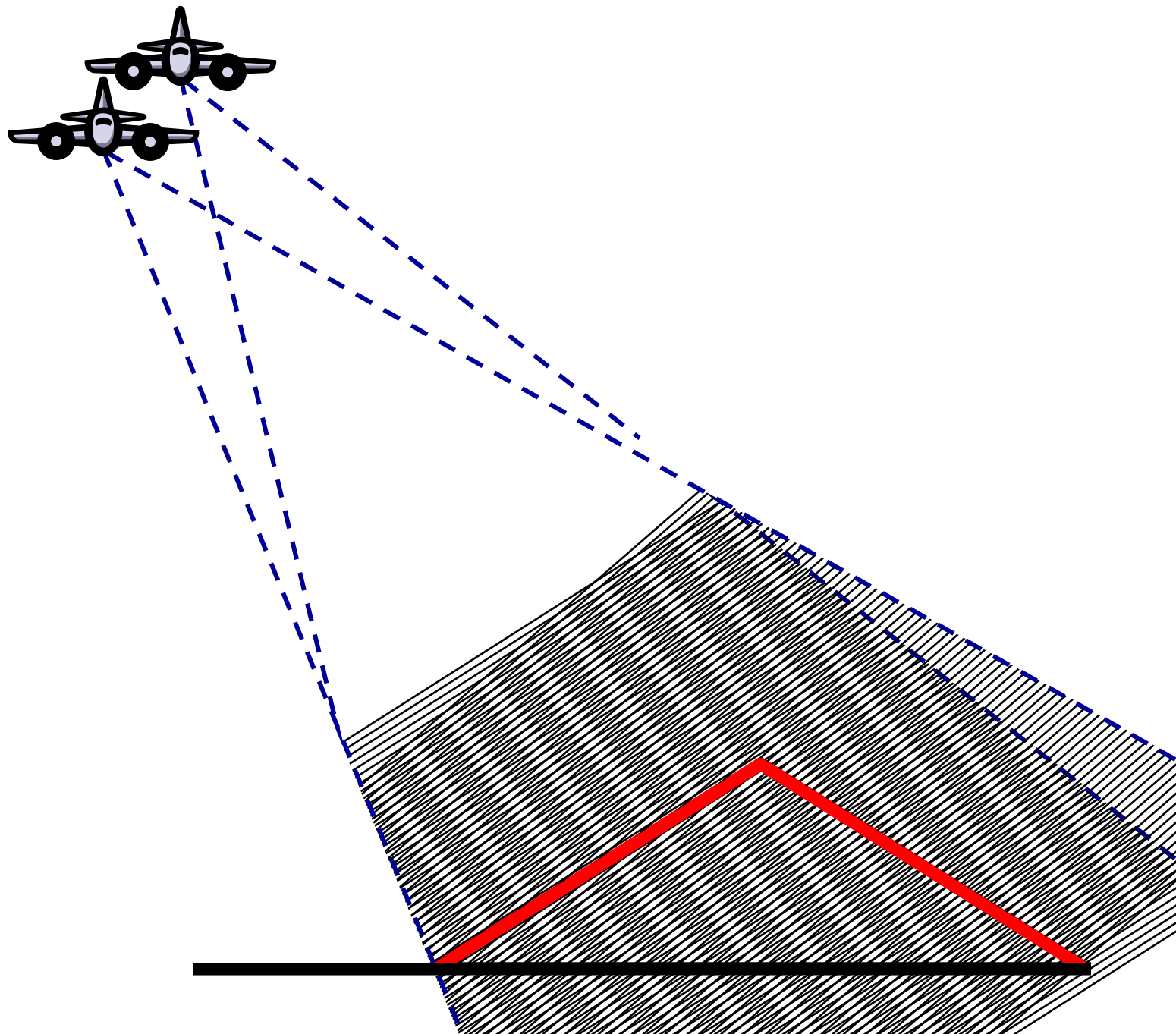
SAR Interferometry (InSAR)

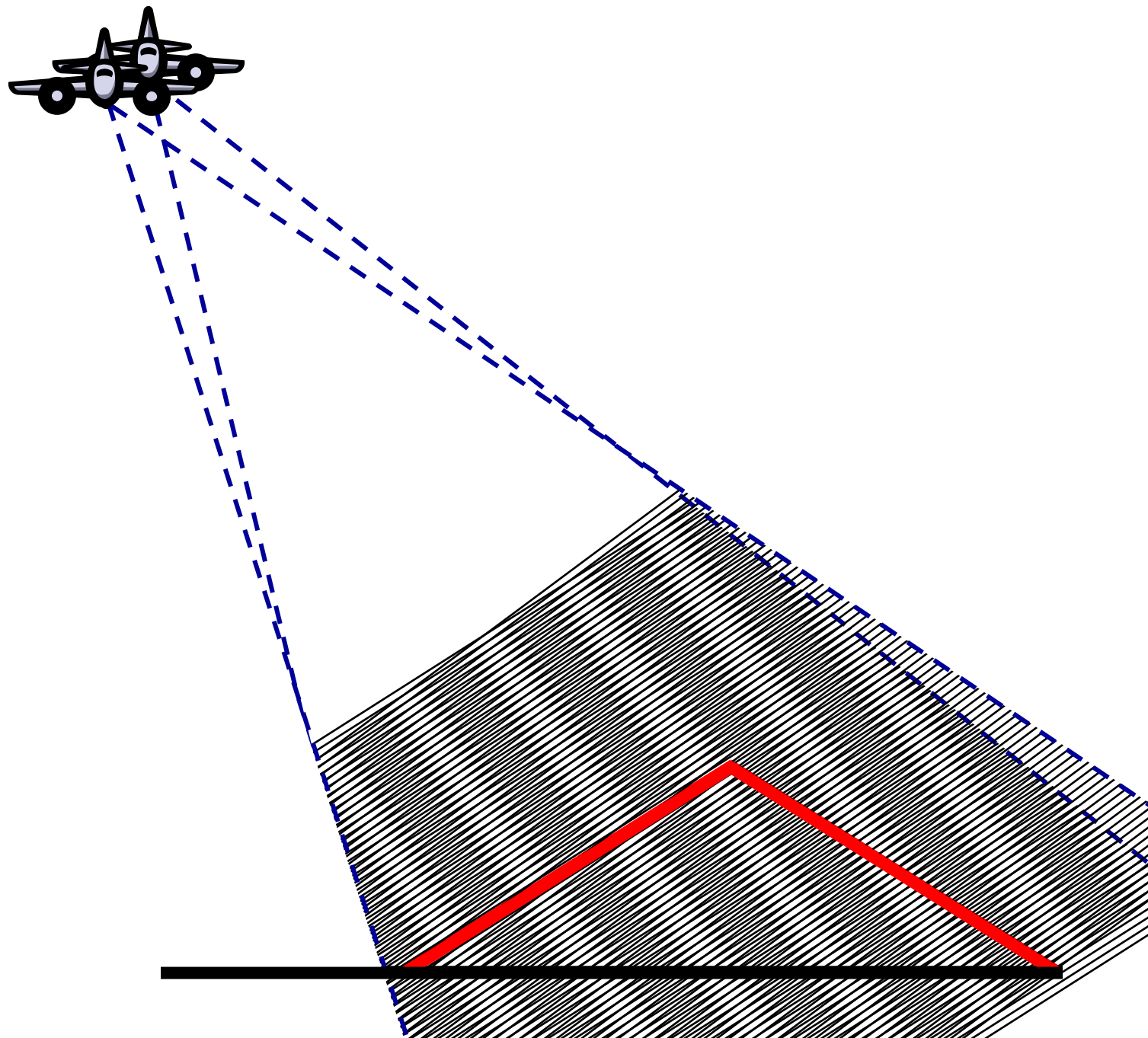


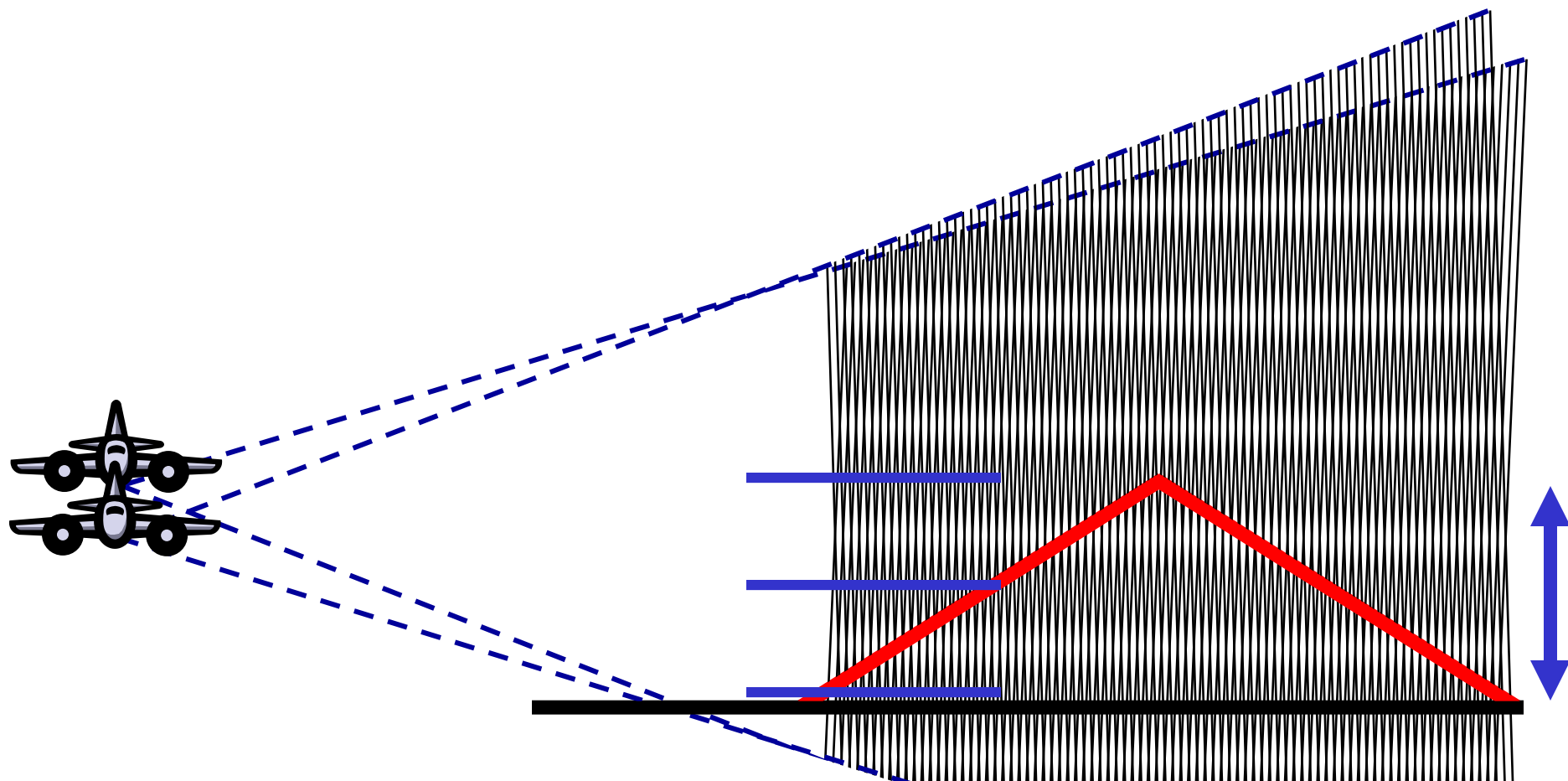




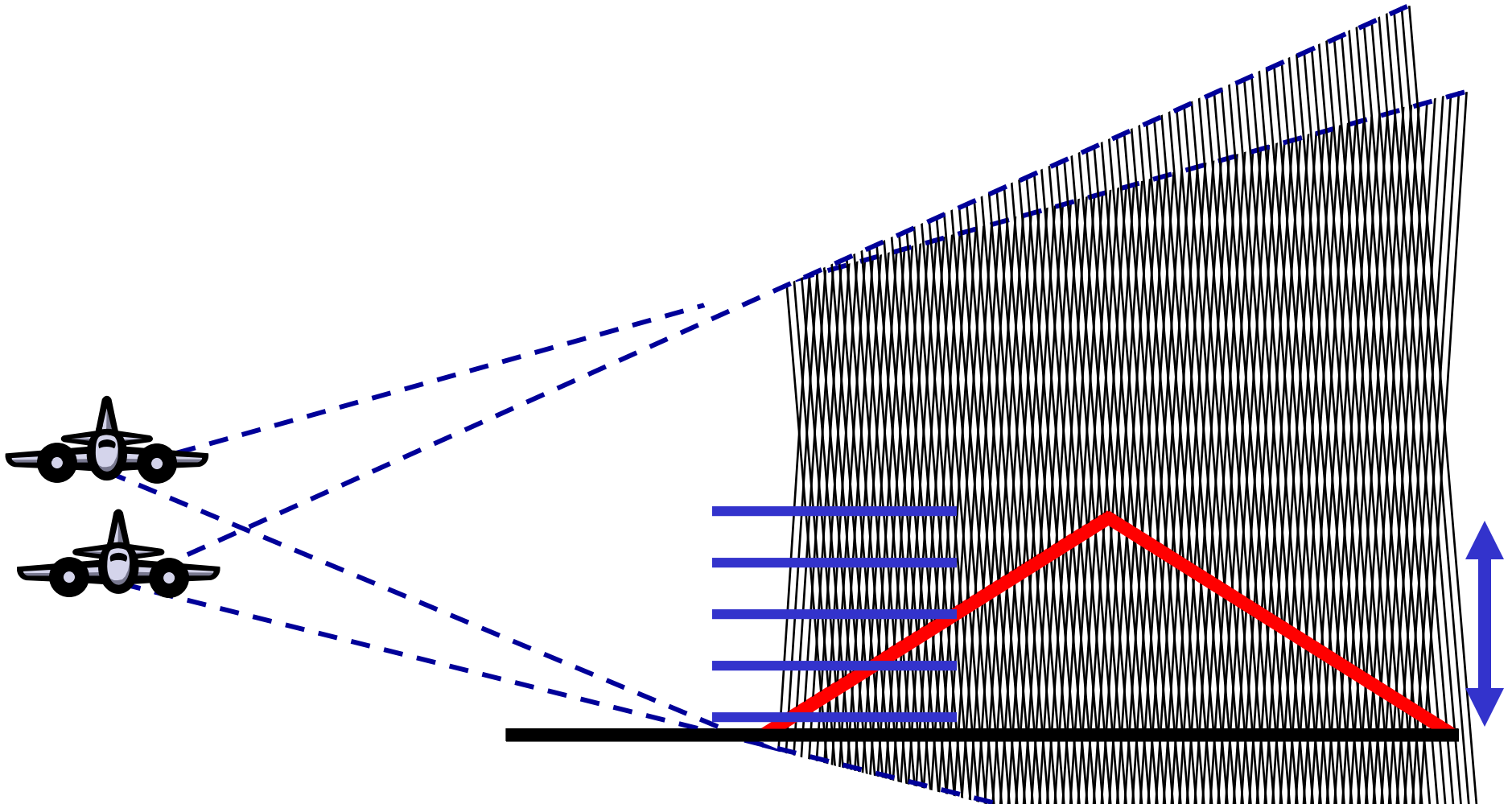








The Phase-to-Height Sensitivity increases with increasing the spatial baseline (i.e. $\Delta\theta$ or B_θ);





Amplitude Images



24 Hours Temporal Baseline

SIR-C / Test Site: Mt. Etna, Italy



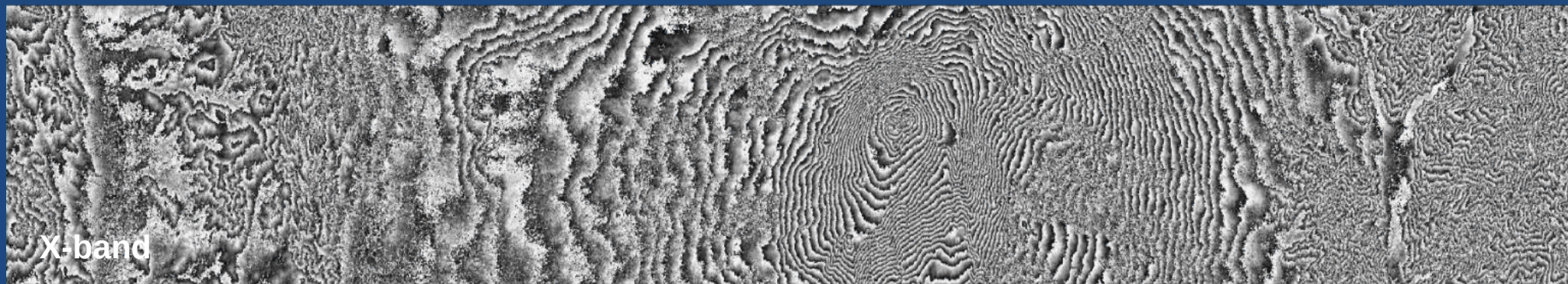
X-band

Phase Images

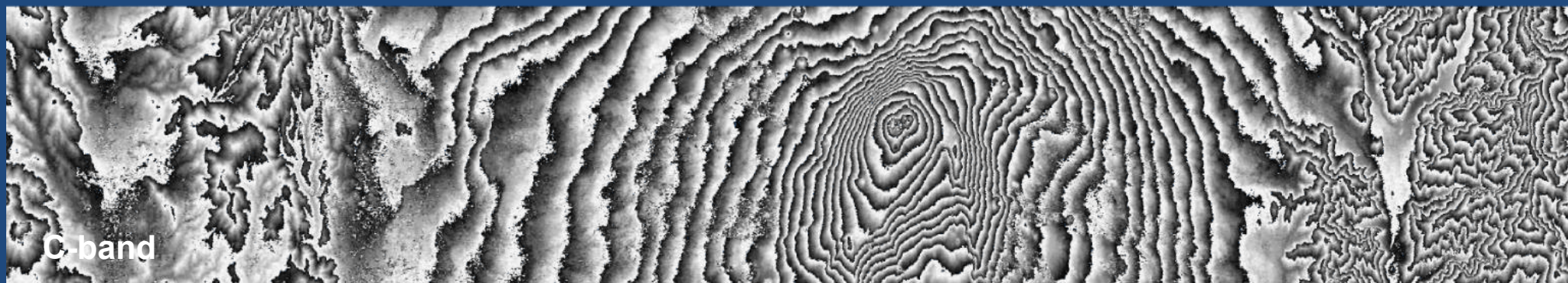
C-band

SIR-C / Test Site: Mt. Etna, Italy

L-band



Phase Images



SIR-C / Test Site: Mt. Etna, Italy



Interferometric Processing

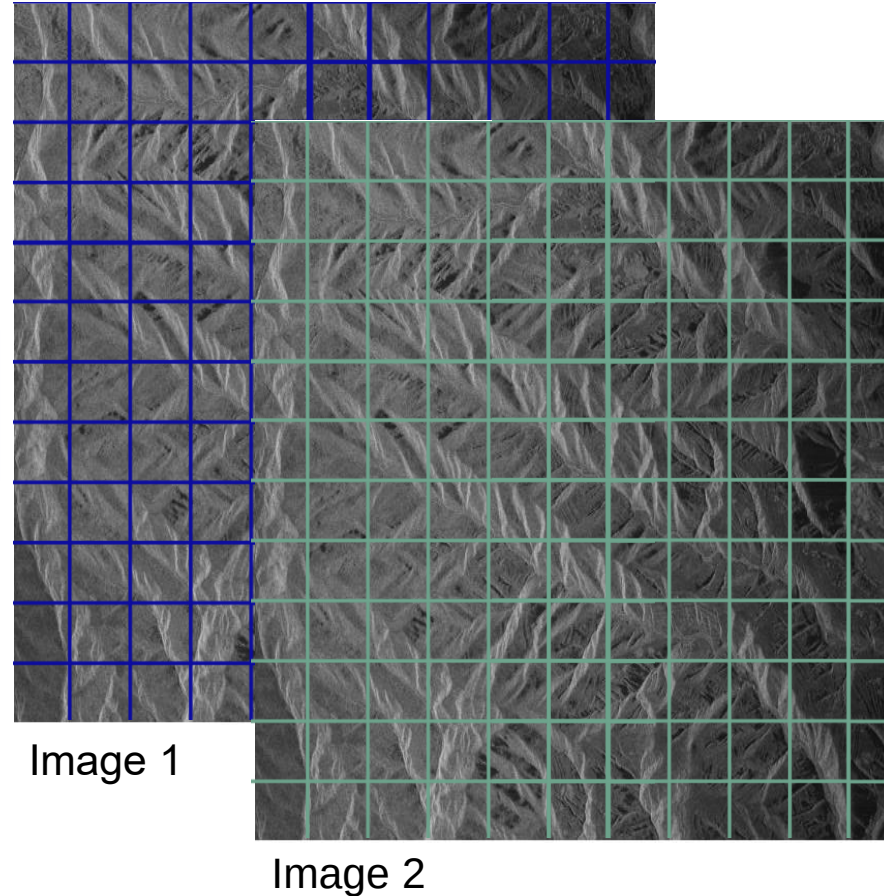
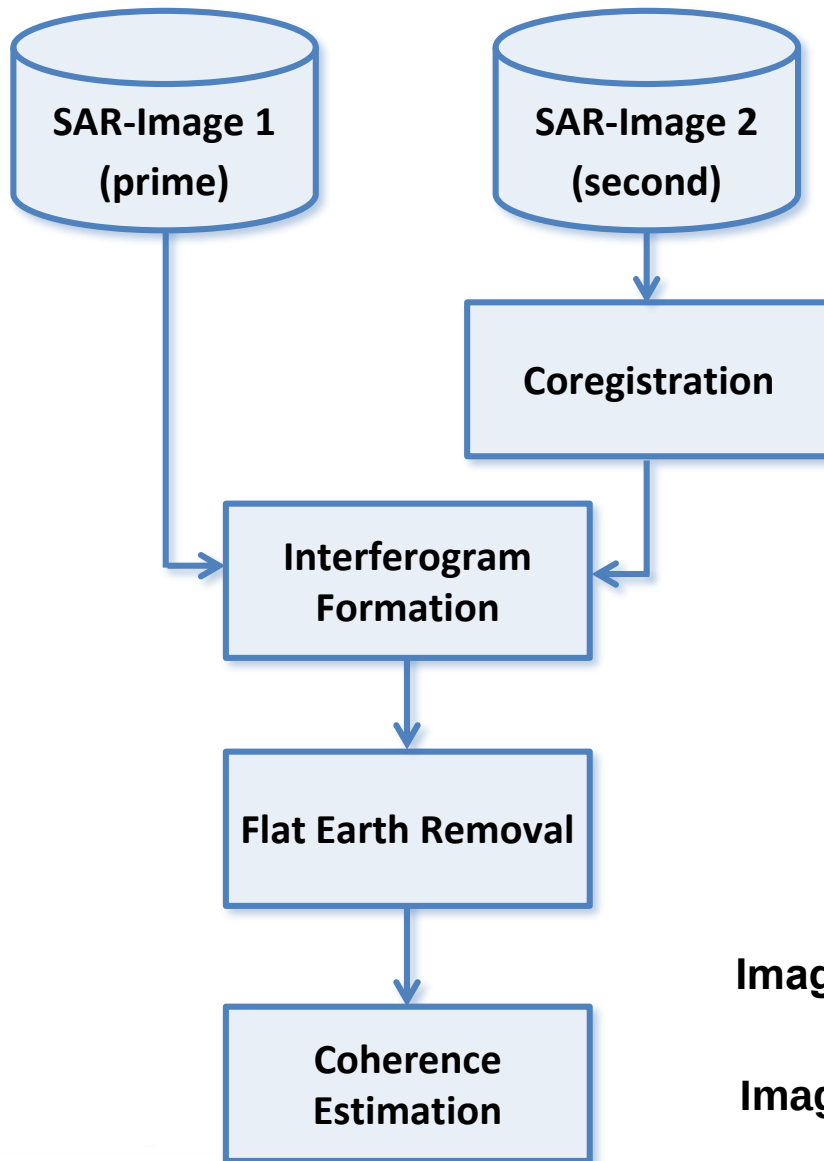
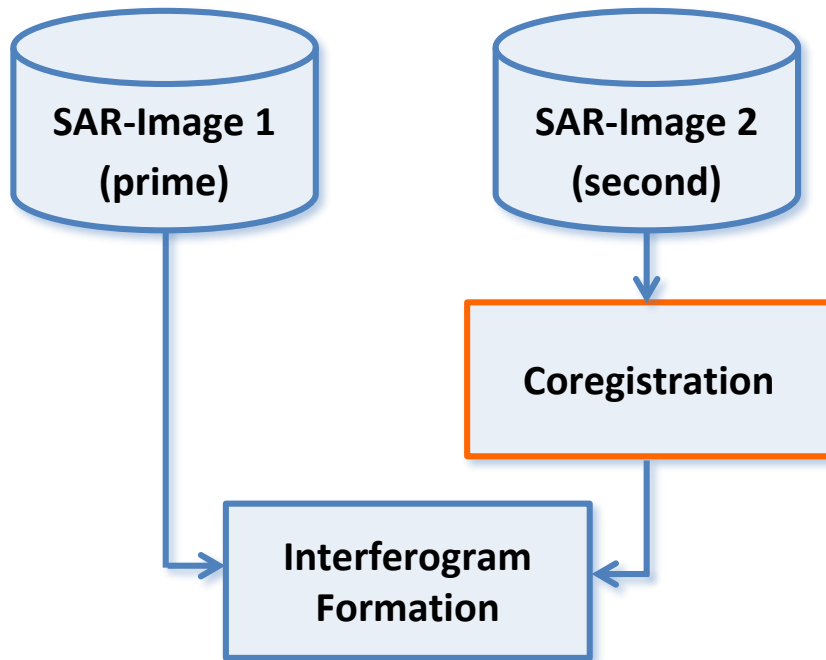


Image 1 (prime): $s_1(x,y) + n_1(x,y)$

Image 2 (second): $s_2 = s_1(x + \Delta x, y + \Delta y) + n_2(x,y)$

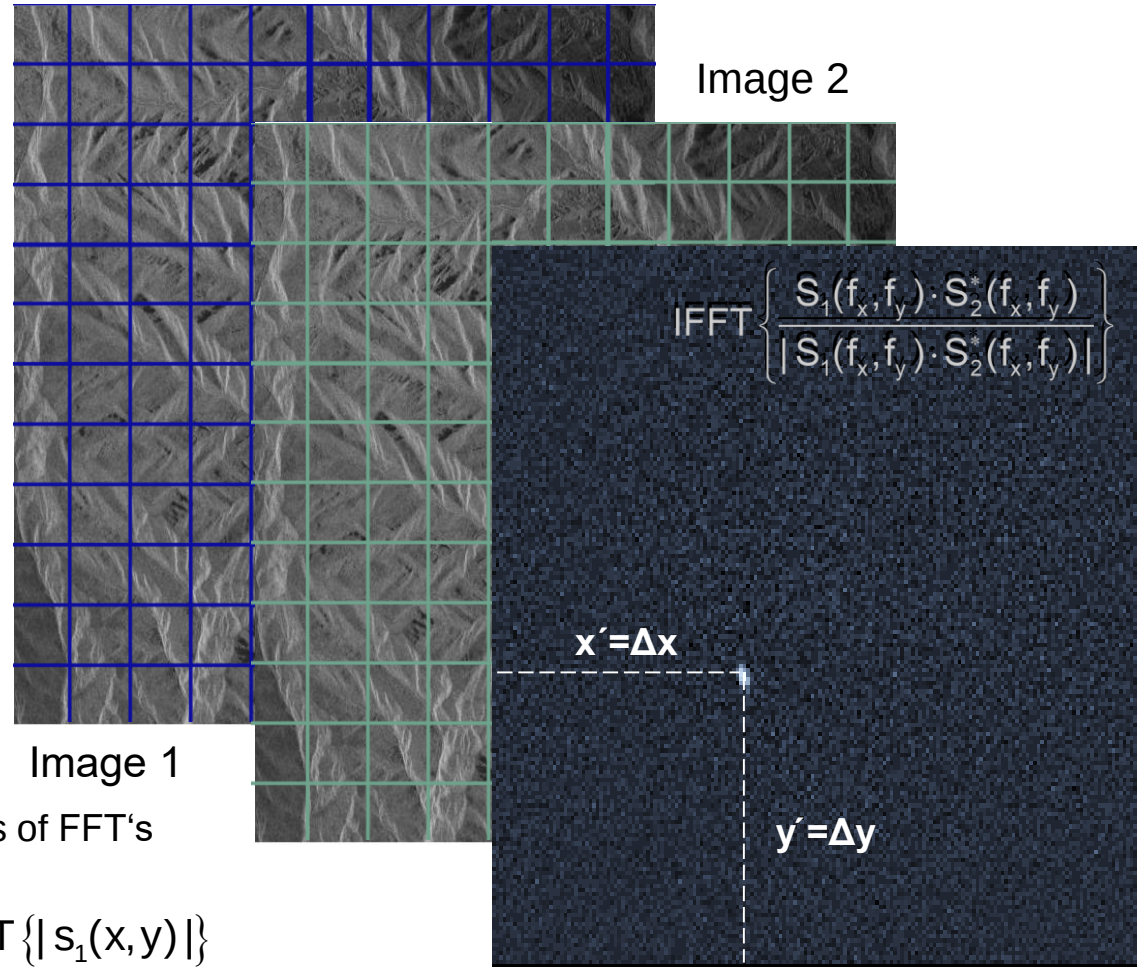
$X_{\text{dim}}, Y_{\text{dim}} := \text{Nr. of image samples in Range, Azimuth}$

Interferometric Processing: Coregistration



Corregistration:= 2D Correlation

implemented by means of FFT's

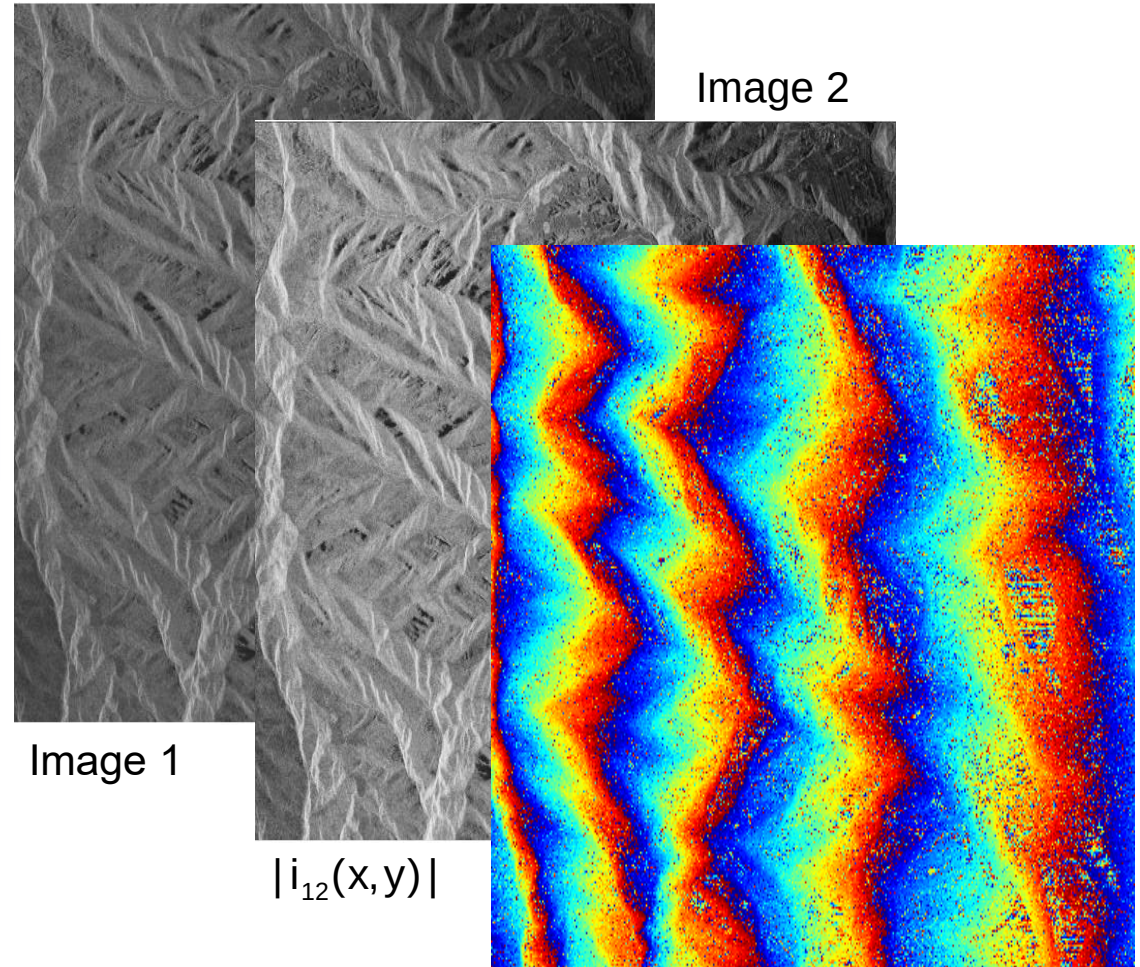
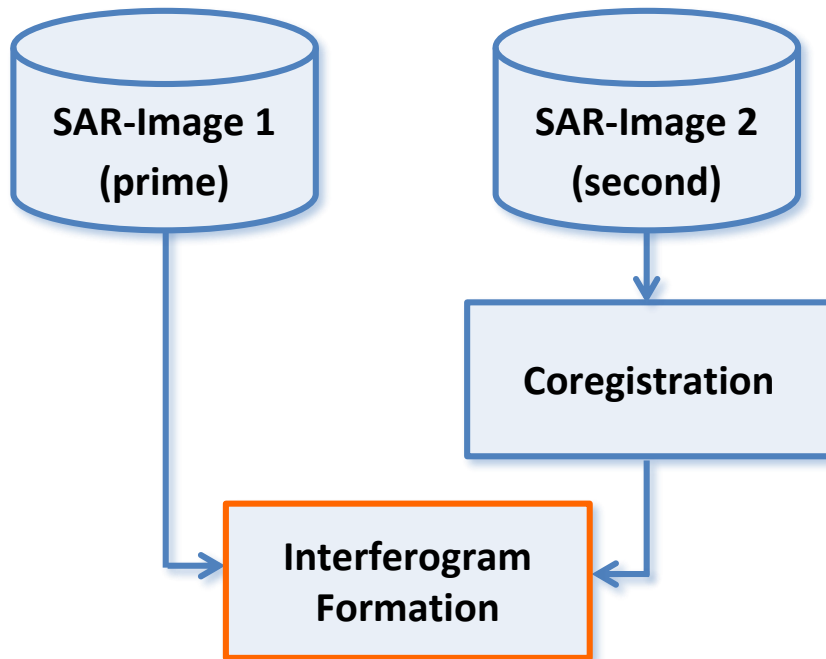


$$s_1(x, y) \quad \bullet \circ \quad S_1(f_x, f_y) = \text{FFT} \{ |s_1(x, y)| \}$$

$$s_2 = s_1(x + \Delta x, y + \Delta y) \quad \bullet \circ \quad S_2(f_x, f_y) = \text{FFT} \{ |s_2(x, y)| \} = S_1(f_x, f_y) \cdot \exp(-i \frac{2\pi}{X_{\text{dim}}} f_x \Delta x) \cdot \exp(-i \frac{2\pi}{Y_{\text{dim}}} f_y \Delta y)$$

$$\max_{x', y'} \left\{ \left| \text{IFFT} \left\{ \frac{S_1(f_x, f_y) \cdot S_2^*(f_x, f_y)}{|S_1(f_x, f_y) \cdot S_2^*(f_x, f_y)|} \right\} \right| \right\} \rightarrow \Delta x = x' \quad \Delta y = y'$$

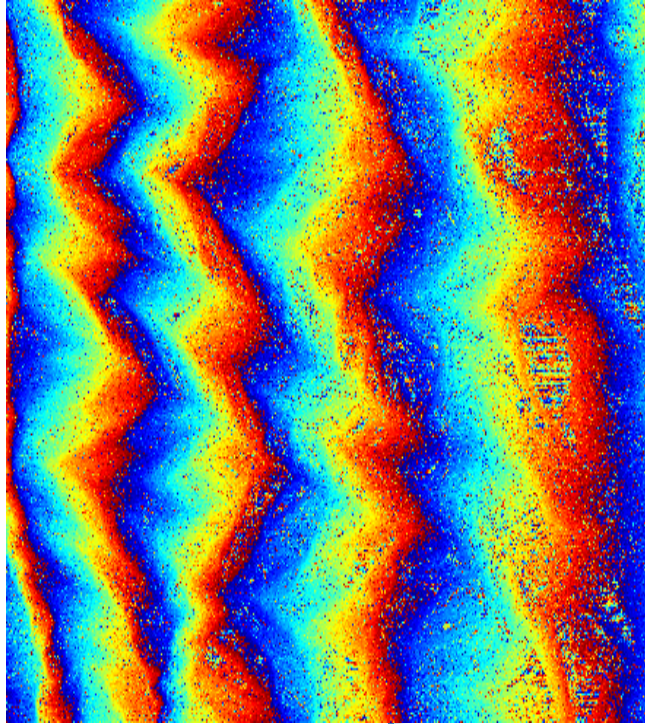
Interferometric Processing: Interferogram Formation



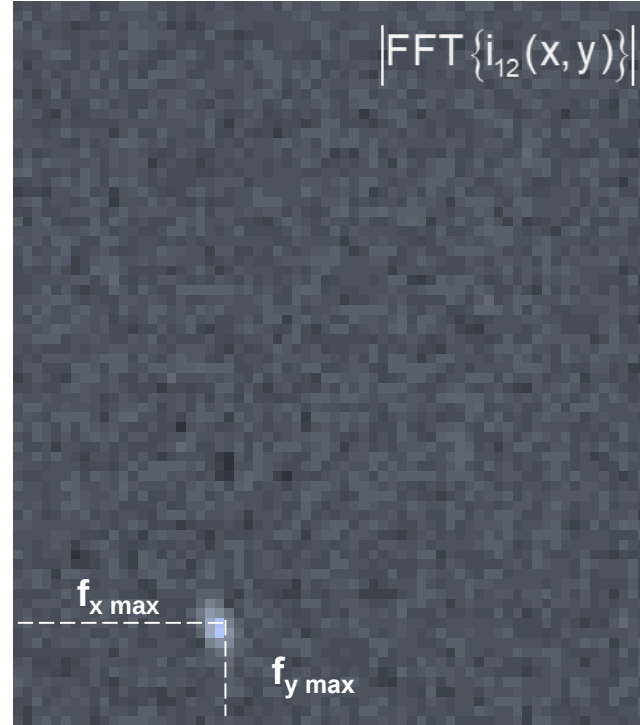
Interferogram Formation: $i_{12}(x,y) = s_1(x,y) \cdot s_2^*(x,y)$

$$\angle i_{12}(x,y) = \arg \{i_{12}(x,y)\}$$

$$\angle i_{12}(x,y) = \arg \{i_{12}(x,y)\}$$



Interferometric Processing: FE Removal



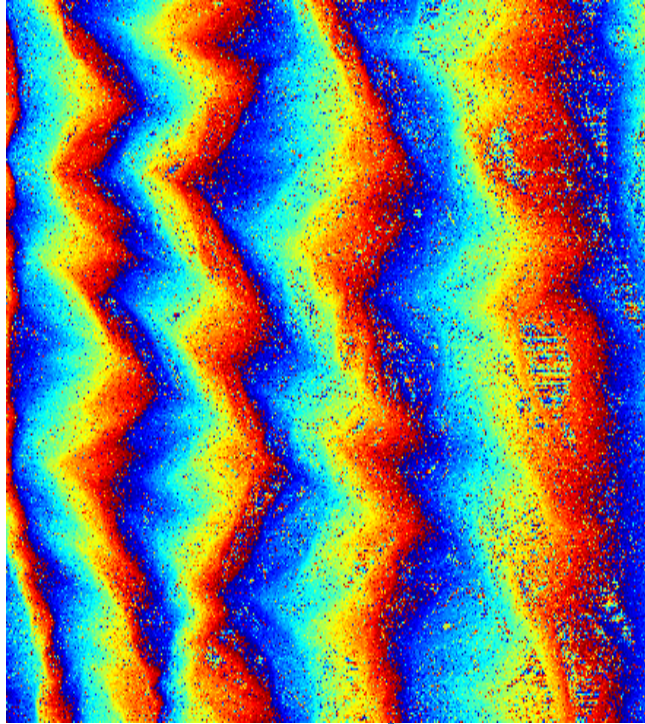
Flat Earth Removal

Coherence
Estimation

Dominant Frequency $f_{\max} = \max_{f_{x\max}, f_{y\max}} \{|\text{FFT}\{i_{12}(x,y)\}|\}$

FE Phase: $\text{FE} = \exp(-i \frac{2\pi}{X_{\text{dim}}} x f_{x\max}) \cdot \exp(-i \frac{2\pi}{Y_{\text{dim}}} y f_{y\max})$

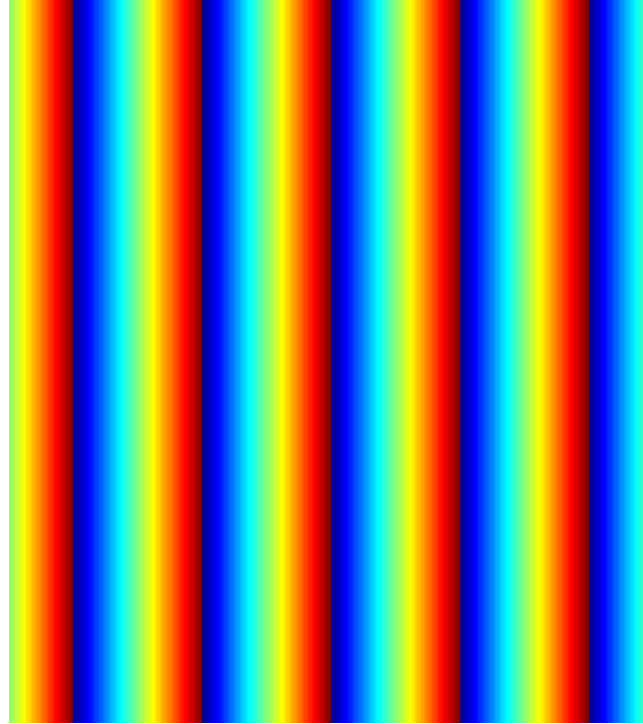
$$\angle i_{12}(x,y) = \arg \{i_{12}(x,y)\}$$



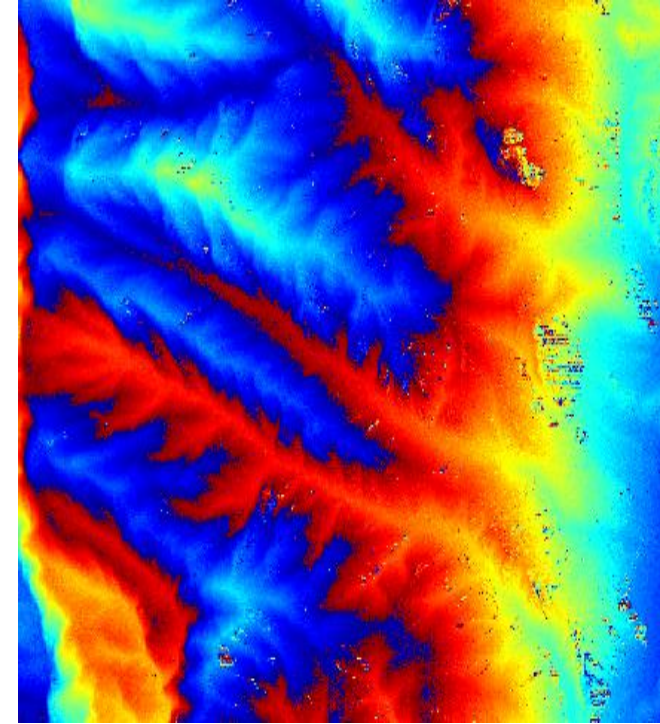
Flat Earth Removal

Coherence
Estimation

Interferometric Processing: FE Removal



$$\angle i_{12}(FE) = \arg \{i_{12}(FE)\}$$



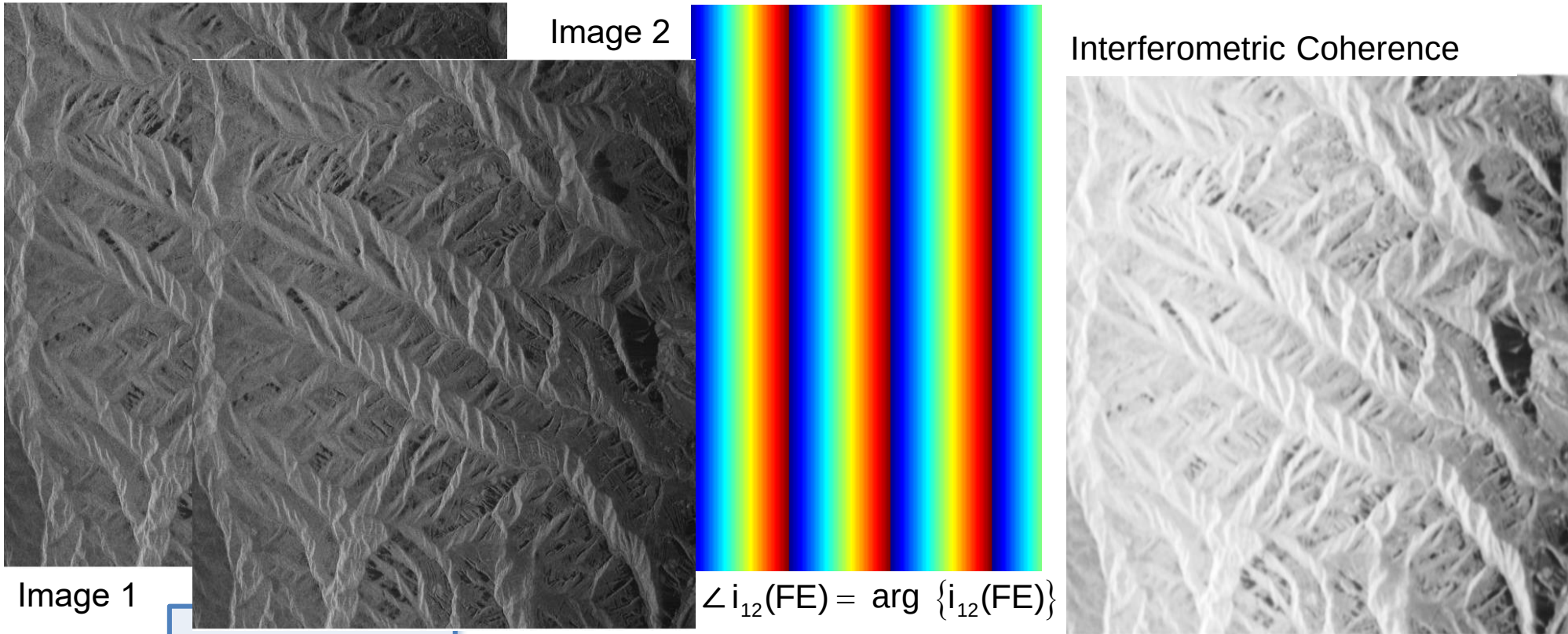
$$\angle i(x,y) = \arg \{i(x,y)\}$$

Dominant Frequency $f_{\max} = \max_{f_{x\max}, f_{y\max}} \{|\text{FFT}\{i_{12}(x,y)\}|\}$

FE Phase: $FE = \exp(-i \frac{2\pi}{X_{\text{dim}}} x f_{x\max}) \cdot \exp(-i \frac{2\pi}{Y_{\text{dim}}} y f_{y\max})$

FE Removal: $i(x,y) = i_{12}(x,y) * FE = s_1(x,y) \cdot s_2^*(x,y) * FE^*$

Interferometric Processing: Coherence Estimation



Flat Earth Removal

Int. Coherence:

$$\tilde{\gamma} = \frac{E\{s_1 s_2^*\}}{\sqrt{E\{|s_1|^2\}} \sqrt{E\{|s_2|^2\}}} = \frac{E\{s_1 s_2^*\}}{\sqrt{E\{s_1 s_1^*\}} \sqrt{E\{s_2 s_2^*\}}}$$

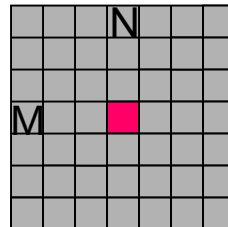
Normalised complex correlation coefficient with $0 \leq |\tilde{\gamma}| \leq 1$

Coherence Estimation

Estimator:

$$\tilde{\gamma} = \frac{\sum_W s_1[i, j] s_2^*[i, j] FE[i, j]^*}{\sqrt{\sum_W |s_1[i, j]|^2} \sqrt{\sum_W |s_2[i, j]|^2}} = \frac{\langle s_1 s_2^* FE^* \rangle}{\sqrt{\langle s_1 s_1^* \rangle} \sqrt{\langle s_2 s_2^* \rangle}}$$

(N x M = W Window)



The Course

Synthetic Aperture Radar (SAR)

Part 3: SAR Interferometry

Prepared by DLR-HR's Pol-InSAR Team

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